Distorted Visual Sequence Pattern Recognition using Deep Learning

Overview

Many modern digital systems need to read text from images that are unclear, noisy, or intentionally distorted. In real-world situations, characters may overlap with each other, contain random noise, be partially hidden, or appear in unusual shapes and positions. These challenges make normal text recognition systems unreliable.

In this challenge, participants must build a deep learning model that can correctly recognize and reconstruct text sequences from visually distorted grayscale images.

The dataset contains grayscale images along with their correct text labels for training. Each image contains a sequence of characters affected by different types of distortions such as:

- Background noise
- Overlapping symbols
- Blur and visual artifacts
- Shape deformation
- Occlusion and random patches
- Irregular spacing and alignment

The goal is to design a model that can learn meaningful visual patterns and accurately predict the correct ordered sequence of characters even under difficult conditions.

Objective

Given a distorted grayscale image containing a hidden text sequence, the model must predict the correct sequence of characters present in the image.

Example:

Input Image → Distorted sequence image

Expected Output:
AXU323

Dataset Structure

The dataset is divided into training and testing sets.

Training Set

The training set contains:

- Input images

- Corresponding text labels

Test Set

The test set contains:

- Only input images

Participants must generate predictions for the test images and submit the predicted text sequences.

Approach

This challenge focuses on building a deep learning system that can accurately recognize and reconstruct text sequences from noisy and distorted images. Participants must design models capable of extracting useful visual features, handling overlapping and corrupted characters, learning sequential patterns, and generalizing well to unseen image distortions while efficiently decoding the correct character sequence.

Participants may use deep learning approaches such as CNNs for visual feature extraction and sequence models like LSTM, BiLSTM, or CRNN for character sequence prediction. Attention-based and transformer architectures may also be explored. Image preprocessing and augmentation techniques can further improve robustness against noisy and distorted inputs.

Evaluation Metric

Submissions will be evaluated using **Character Error Rate (CER)**, which measures how different the predicted character sequence is from the actual target sequence. CER is based on Levenshtein Distance and calculates the minimum number of insertions, deletions, and substitutions required to transform the predicted text into the correct text. Since the comparison is performed sequentially, incorrect character ordering also increases the CER score. A lower CER indicates better model performance.

Evaluation process will also consider the overall approach, experimentation, pipeline, model design, and the quality of the submitted notebook. Participants are encouraged to demonstrate genuine problem solving efforts and meaningful understanding rather than shallow or copy pasted GPT solutions.

Submission Guidelines

Notebook (.ipynb)

Participants are required to submit a well documented and reproducible notebook explaining the complete workflow of their solution. The notebook should clearly explain the methodology, model architecture, training strategy, evaluation, outputs etc.

Prediction File (.csv)

The submission file should contain the image filename and the corresponding predicted text sequence for each test sample.

File Name: submission_<name>_<enroll no.>.csv

Format:

image,prediction
test-0.png,AXU323
test-1.png,BT91KD
test-2.png,QWER12

Dataset Link : -

https://drive.google.com/drive/folders/1lRUA-1uCCXfks8kpypFV-4f0UepWoLkU?usp=sharing